Shifting Dynamics in Bank Funding of NBFIs: The Rise of Credit Lines

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Introduction

We examine the shifting dynamics in bank funding of nonbank financial institutions (NBFIs) from 1980 to 2024. We **document three key facts**:

1. **NBFIs' liabilities have exponentially increased since 1980**, from which **bank funding** has been relatively stable accounting for **3.5% of total liabilities**, on average;
2. Since 2012, **on-balance-sheet funding by banks** has declined sharply **from 4.7% to 2.5%**;
3. **Credit lines** extended to NBFIs have **increased from 0.7% to 1.0%** of total NBFI liabilities, highlighting a recent shift in how banks fund NBFIs.

While reducing on-balance-sheet exposure to NBFIs might enhance financial stability, the growing reliance on credit lines, now accounting for approximately 2% to 3% of GDP, has increasingly introduced systemic vulnerabilities that can be amplified during periods of financial stress.

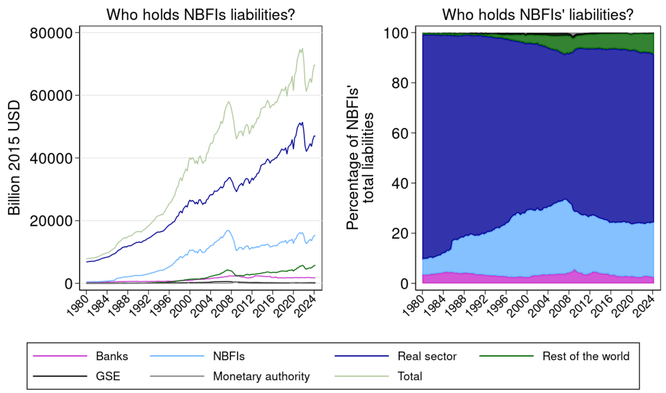
Data

- **Federal Reserve System Enhanced Financial Accounts (From Whom to Whom)**: Tracks asset and liability flows between major sectors of the economy including banks and NBFIs over the period starting 1980 and ending 2024;
- **Form Y-14Q Schedule H.1**: Accurately captures credit and credit lines provided by banks to NBFIs between 2012 and 2024;
- **Form FR Y-9C Schedule HC-L**: Allows us to estimate credit lines back to 1990 using "Other unused commitments" as a proxy.

Banks' holdings of NBFIs' liabilities

Funding by banks to NBFIs has remained relatively small and stable over the period 1980–2024 at an average of 3.5% of total NBFIs' liabilities. This is much less than the funding provided by the Real Sector and other NBFIs.

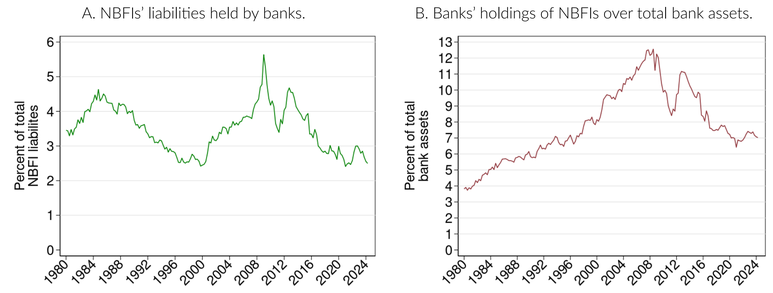
Figure 1. Comparison of funding to NBFIs in inflation-adjusted and relative terms (1980–2024).



Zooming in on bank funding of NBFIs

- Bank funding to NBFIs has remained low in proportional terms. The share of **bank holdings of NBFI liabilities relative to total NBFI liabilities** ranged between 2% and 6% since 1980;
- The share of **bank holdings of NBFI liabilities relative to total bank assets** displays a steady increase from 4% in 1980 to 12.5% in 2008, followed by a decrease to 7% in 2024.

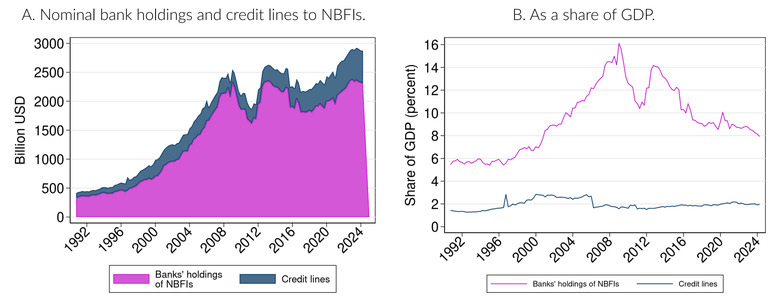
Figure 2. Bank funding of NBFIs (1990–2024).



The growing role of credit lines

- **Credit lines** are contingent liabilities that remain off banks balance sheet until drawn. While traditional measures of bank funding to NBFIs, such as holdings of liabilities, suggest a stabilization in nominal terms since 2008, this view neglects the **growing importance of credit lines**, which increased from USD 0.3 trillion in Q1 2008 to USD 0.6 trillion as of Q4 2024;
- According to **Y-9C data**, credit lines, proxied by other unused commitments, currently account for 2% of GDP, **in line with its historic mean**, which is measured as 1.9%;
- When using **Y-14 data** to capture more precisely credit relationships between banks and firms, we uncover that total credit lines to NBFIs amount to **almost 3% of GDP**.

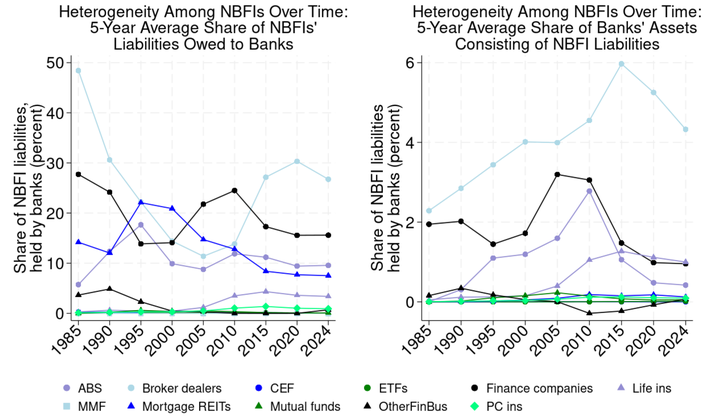
Figure 3. The silent role of credit lines in addition to bank holdings of NBFIs (Y-9C data: 1990–2024).



Heterogeneity Among NBFIs Over Time

- We document **substantial heterogeneity across subsectors**. Mutual funds, Money Market Funds and PC Insurance companies do not significantly rely on bank funding;
- **Broker-dealers** have historically been the **most interconnected subsector** followed by Finance Companies;
- Over the last decade, **bank funding has decreased across all subsectors**.

Figure 4. Bank funding across NBFI subsectors.



Conclusion

- The importance of bank funding to NBFIs is relatively small (Figure 1) and has been decreasing since 2012 (Figure 2);
- While on-balance-sheet funding by banks has declined as a share of GDP since 2012, the use of credit lines has expanded, now representing approximately 2% of GDP (Figure 3);
- The expansion of credit lines as contingent liabilities increases banks' potential exposure to liquidity shocks during periods of stress;
- Heterogeneity within the NBFI sector is significant and time varying (Figure 4); it can be taken into account when designing regulation.

References

[1] Viral V. Acharya, Nicola Cetorelli, and Bruce Tuckman. Where do banks end and nbfis begin? *Federal Reserve Bank of New York Staff Reports*, (1119), 2024. URL `https://www.newyorkfed.org/research/staff_reports/sr1119.html`.

[2] Board of Governors of the Federal Reserve System. Financial stability report. November 2024. URL `https://www.federalreserve.gov/publications/files/financial-stability-report-20241122.pdf`.

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